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To cite this article: Yongyang Xu, Zhanlong Chen, Zhong Xie & Liang Wu (2017) Quality assessment of building footprint data using a deep autoencoder network, International Journal of Geographical Information Science, 31:10, 1929-1951, DOI: [10.1080/13658816.2017.1341632](https://doi.org/10.1080/13658816.2017.1341632)

To link to this article: <https://doi.org/10.1080/13658816.2017.1341632>



Published online: 19 Jun 2017.



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Quality assessment of building footprint data using a deep autoencoder network

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ABSTRACT

Volunteered geographic information (VGI), OpenStreetMap (OSM), has been used in many applications, especially when official spatial data are unavailable or outdated. However, the quality of VGI remains a valid concern. In this paper, we use the matched results between OSM building footprints and official data as the samples for training an autoencoder network, which encodes and reconstructs the sample populations according to unknown complex multivariate probability distributions. Then, the OSM data are assessed based on the theory that small probability samples contribute little to the autoencoder network and that they can be recognized by the higher reconstructed errors during training. In the method described here, the selected measures, including data completeness, positional accuracy, shape accuracy, semantic accuracy and orientation consistency between OSM and official data, are used as the inputs for a deep autoencoder network. Finally, building footprint data from Toronto, Canada, are evaluated, and experiments show that the proposed method can assess the OSM data comprehensively, objectively and accurately.

ARTICLE HISTORY

Received 8 December 2016

Accepted 8 June 2017

KEYWORDS

Autoencoder network;
OpenStreetMap; quality
assessment; building
footprint

1. Introduction

The development of contemporary technical infrastructure, including location-enabled mobile phones and web 2.0, has facilitated collecting and accessing geospatial data. Consequently, volunteered geographic information (VGI; Goodchild 2007) has been widely utilized in applications ranging from transportation routing to emergency management (Goodchild and Glennon 2010). As one example of successful and popular VGI projects, OpenStreetMap (OSM) 'has become a potential competitor to public and commercial geodata providers' (Neis *et al.* 2011), and it boasts more than 2.8 million registered members worldwide (OSM 2016).

As a user-generated online map of the world, OSM is editable by registered users, and it is also free, up to date and comprehensive (Flanagin and Metzger 2008, Hashemi and Abbaspour 2015). Moreover, OSM even includes data for relatively less-developed and sparsely populated regions (Forghani and Delavar 2014), which makes it a popular and beneficial application for many users. However, the quality and value of OSM on its lack of

coordinated actions and industrial-grade sensing devices has raised concerns, particularly because that its contributors may be untrained and unqualified. Reliability is a necessity for applications that utilize geographic data; thus, the application of OSM across a variety of fields depends on the assurance of spatial data quality (Goodchild and Li 2012). Consequently, the proper evaluation of data quality provided by volunteers is necessary before OSM data are used for research or other specific purposes. To this end, several studies have investigated measures to determine the quality of OSM data (Girres and Touya 2010, Ludwig *et al.* 2011, Al-Bakri and Fairbairn 2012, Corcoran and Mooney 2013, Forghani and Delavar 2014, Fan *et al.* 2016). In the past, data quality has been assessed using the relationships of features between OSM data and official data. For example, the quality of OSM data has been assessed by comparing it with an ordnance survey (OS) data set and then calculating the positional accuracy and completeness (Haklay 2010). Directional relationships, topological relationships and metric distance relationships are employed to assess the logical consistency of OSM data (Hashemi and Abbaspour 2015), whereas hierarchical consistency analyses and classification plausibility are utilized to assess the integrity of OSM data (Ali and Schmid 2014). Previous researchers have provided useful insights into the various methods that can be used to evaluate OSM data. However, these methods assess the quality of the data either by separately calculating the quality assessment measures, such as completeness, semantic accuracy, positional accuracy and shape accuracy, or by combining certain criteria, including the semantic and structural similarities between OSM and an official data set, with weights provided by experts or experienced researchers (Al-Bakri and Fairbairn 2012). These approaches lack a comprehensive and objective assessment of OSM data, and fail to provide a single assessment measure to incorporate all of the assessment measures. Accordingly, a comprehensive assessment method is desirable to determine which data are superior among the data with different advantages.

Data quality is an expression of the data features, and it can be analysed via the deeper level law of the data using self-learning. As a consequence of their unsupervised learning capabilities, self-learning systems are widely used throughout particular fields for anomalous data analyses (Fiore *et al.* 2013, Sahasrabudhe and Namboodiri 2013). As a knowledgeable machine learning tool, restricted Boltzmann machines (RBMs) (Murray 2001) have been utilized within many applications, such as the multivariate distribution model and deep-learning construction (Hinton and Salakhutdinov 2006, Hinton 2012). Unfortunately, RBMs are limited to continuous-value and non-Gaussian inputs (Hinton and Salakhutdinov 2006). To overcome these limitations, the continuous restricted Boltzmann machine (CRBM) (Chen and Murray 2003) was developed. CRBMs are trained iteratively by minimizing contrastive divergence (MCD) (Hinton 2002). Deep architecture learning attempts to extract data features by exploiting multiple processing layers as the main category for abstracting spatial objects at a high level (Bengio 2009). Accordingly, the deep autoencoder network is modelled after RBMs. Compared with supervised learning systems, a deep autoencoder network as the unsupervised learning system learns the data features by analysing the deeper-level law of the data. During training of the model, this network does not require a labelled sample data set (i.e. the ground truth), and the trained model is subsequently evaluated using the reconstruction error. However, the deep autoencoder network is valid only for binary inputs. Fortunately, unlike RBMs, CRBMs can be used to build an autoencoder network when the input is continuous (Valentine and Trappert 2012), and they can be trained by minimizing the

differences between the input and output data (Hinton and Salakhutdinov 2006). Compared with large probability samples, small probability samples are unlikely to contribute to a deep autoencoder network during model training; therefore, the trained model will perform poorly at encoding and reconstructing small probability samples and will have larger reconstructed errors (Sahasrabudhe and Namboodiri 2013, Chen 2015).

In many circumstances, authoritative data is of worse quality than VGI for older technologies at a smaller scale (1:24,000) (Goodchild and Li 2012). However, because most building footprints (i.e. London, Toronto) are digitized using Bing maps (Goetz and Zipf 2012, Fan *et al.* 2014, OSM 2017a), which is easily distinguished using the source tag (OSM 2017b), and fewer new technologies or Global Positioning System devices are used in comparison with the collection of road networks in OSM, we propose a new method to assess the quality of OSM building footprint data using the deep autoencoder network. This new method first calculates the data completeness, positional accuracy, shape accuracy, semantic accuracy and orientation consistency as the quality assessment measures. Then, those measures are treated as the inputs and then encoded and reconstructed by a trained network. Consequently, anomalously performing OSM data will show a larger reconstructed error which is corresponding to the quality of the OSM data, an anomalous data point that is larger than the average value has better quality in the OSM. This method is capable of achieving a more comprehensive and objective perspective of the quality of geospatial data.

2. Data preprocessing for quality assessment

Data preprocessing is an important prerequisite for data integration and comparison. In this work, the data preprocessing includes two components: the matching of corresponding building footprints in OSM with those in a reference data set and the partitioning of the test data into a number of cells.

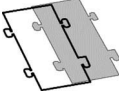
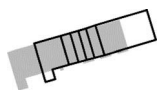
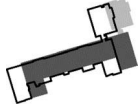
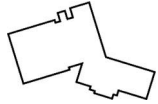
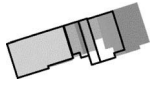
2.1. Matching corresponding building footprints in OSM with those in a reference data set

Most of the building footprints within OSM have been digitized according to Bing map imagery. Therefore, the inherent precision corresponds with the resolution of the Bing images, and several buildings may be digitized as a footprint in OSM. Many such cases are observed, including 1:m, n:1 and n:m, which are shown in Table 1.

With the building footprints in OSM based on Bing map imagery, the corresponding pairs of higher buildings in OSM and the reference data set might demonstrate a larger offset, and the area where two large buildings overlap in an oblique image will exhibit a larger area percentage. Because the resolution of Bing map imagery may affect the quality of OSM data, two or more buildings can be digitized as one building, and one building in a Bing map may be digitized into several buildings in OSM. For these reasons, the matching of corresponding buildings is a two-stage process that includes filtering and refinement.

In the filtering stage, an overlapping method is introduced based on the area of overlap between a building footprint and its neighbours in another data set that are less

Table 1. Footprint-to-footprint cardinalities in the OSM and the reference map and their respective illustrations.

1:1	1:0	1:m
		
Footprint in the OSM corresponds to one footprint in the reference map	Footprint in the OSM corresponds to no footprint in the reference map	Footprint in the OSM corresponds to several (m) footprints in the reference map
$n:1$	0:1	$n:m$
		
Several (n) footprints in the OSM correspond to single footprint in the reference map	No footprint in the OSM corresponds with any footprint in the reference map	Several footprints (n) in the OSM correspond to several (m) footprints in the reference map

than a threshold T_0 (Rutzinger *et al.* 2009). If one building footprint in OSM matches with a footprint in the reference data set, the following criterion will be met:

$$\frac{\text{Area}_{\text{overlap}}}{\min(\text{Area}_{\text{OSM}}, \text{Area}_{\text{ref}})} > T_0 \tag{1}$$

OSM building footprints represent a simplified version of the reference data. Most of the buildings are matched well, and the matched results have been analysed via experiments. However, a building footprint in one data set may correspond to two separate footprints in another data set (Figure 1).

In Figure 1(a), footprint A obviously matches with footprint B, whereas in Figure 1(b) footprint A corresponds to both footprints B and C; however, both of them are regarded

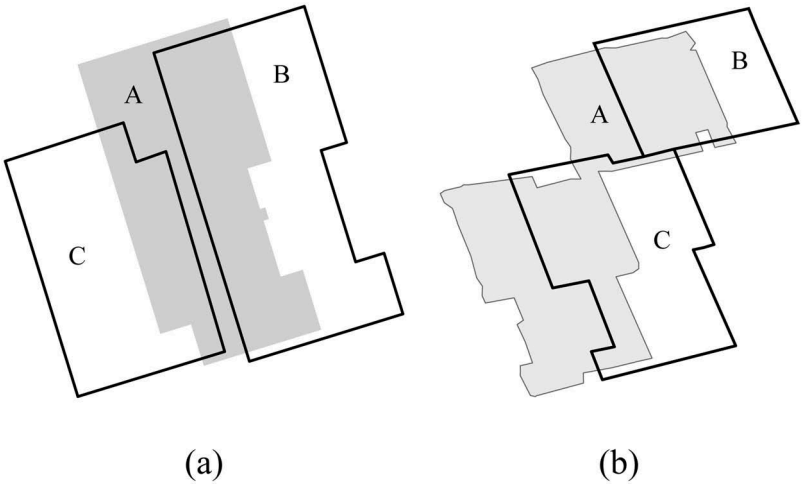


Figure 1. Footprint A intersects two footprints B and C.

as 1: n by the overlapping method. This problem is resolved in a refinement step by adding the area of polygons in the two data sets. If the difference of the added areas is smaller than a threshold T_0 (the same as that in formula (1)), then they matched as a relationship 1: n . Otherwise, the shape similarity will be calculated to identify the corresponding building pairs. Using [Figure 1](#) as an example, if we consider

$$\frac{(\text{Area}_B + \text{Area}_C) - \text{Area}_A}{\text{Area}_A} < T_0 \quad (2)$$

then footprint A will match footprints B and C . If not, then the shape similarity (sim_{AB}) between footprint A and footprint B and the similarity (sim_{AC}) between footprint A and footprint C will be calculated. Footprint A matches with footprint B when sim_{AB} is larger than sim_{AC} .

2.2. Dividing the map into cells

It is necessary to divide the research area into regular raster data, such as squares, triangles or hexagons, while evaluating the data quality so that every cell is treated as a sample. This process allows the deep autoencoder network to learn the data features and analyse the quality of the building footprints, which increases the number of samples and avoids the overfitting of the deep learning. Moreover, partitioning the research area could substantially reduce the computation during the search for corresponding footprints within the different data sets (Haklay 2010, Fan *et al.* 2014, Forghani and Delavar 2014). OSM represents a type of VGI that is produced by a lot of volunteers, and a single person may only contribute a piece of data to OSM. Thus, assessing quality in cells is more effective than the entire data as a whole. Different division techniques may lead to different samples, which will affect the details of the assessment results. A smaller cell size will produce more detailed assessment results and lead to a larger number of samples, which could train a more stable model. Hexagonal cells provide for more complete coverage of the study area relative to squares or triangles (Roick *et al.* 2011); this shape simultaneously shares a greater number of edges with neighbouring cells, which will produce a smoother assessment result. The different results obtained using the division methods are shown ([Figure 13](#)) experimentally.

In case that a building is split into several parts to generate multiple mismatches, a building footprint belongs to a cell only when its centroid is located in the cell. In addition, if the centroid is located within an overlap of the division lines, the building is allocated to the cell on the left or above the centroid. To ensure that the corresponding buildings are divided into the same cell, neighbouring cells are included in a buffer to identify matching pairs. For instance, building a belongs to the current cell A , and eight neighbouring cells are used to identify the corresponding building a' ([Figure 2](#)) using the method discussed in [Section 2.1](#).

If the separated parts of a building are allocated into different cells according to the division line, then the problem is resolved with two approaches ([Figure 3](#)). If the separated parts of a building correspond to a combined building in the other data set ([Figure 3\(a\)](#)), then those separated parts will be adjusted to the same cell based on the corresponding combined structural footprint. Otherwise, the separated parts in one data set that correspond to the separated parts in the other data set ([Figure 3\(b\)](#)) will be treated as a single footprint for computation because they are physically separated.

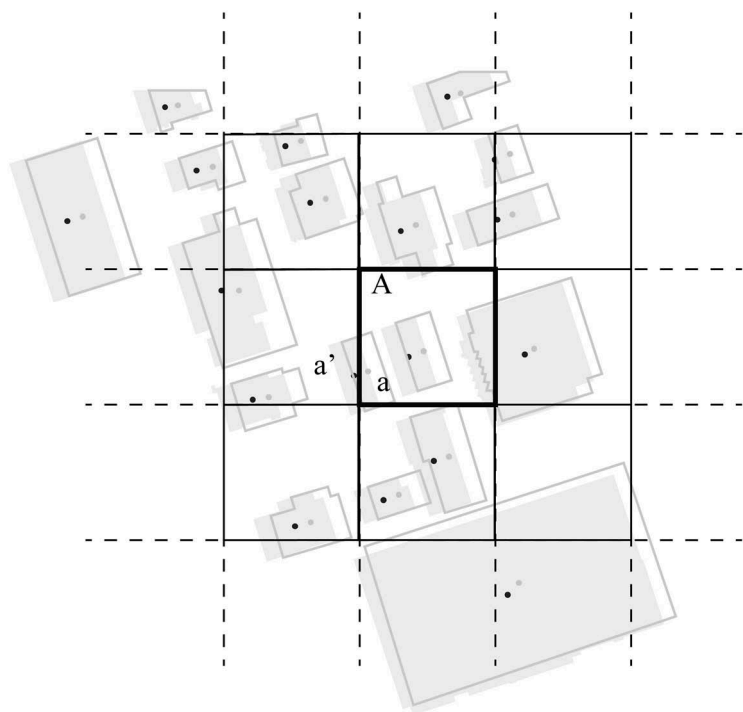


Figure 2. Partitioned map and matching of a pair of building footprints.

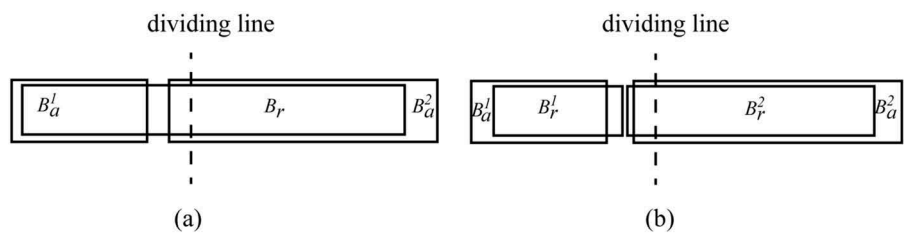


Figure 3. Separated parts of a building that are partitioned by a dividing line.

3. The selected measures for the quality of building footprints in OSM

Some measures should be calculated as input data for the deep autoencoder network so that the quality of the OSM can be assessed based on the reconstruction errors. However, none of the quality assessment measures alone, such as the completeness, semantic accuracy, positional accuracy or shape accuracy, is of particular use for the assessment model. In addition, each dimensional unit of input data must be independent in the deep autoencoder network. Therefore, to assess the quality more comprehensively and identifiably, several measures are utilized in this work, and they are described below.

3.1. Data completeness

This variable is a measure of missing data and describes data that are expected to be observed or data that should be included in the data set but are unexpectedly missing. Values of the true positive (TP) and false negative (FN) are introduced in this paper. TPs denote the matched footprints between OSM and the reference data set. FNs are footprint areas belonging to the reference data set but not to OSM.

$$\text{Completeness} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

3.2. Positional accuracy

This variable is used to evaluate the error in the geographical location of a building in OSM relative to the actual building location. In this method, the centroids of building footprints in each matched pair are calculated initially both when they fall inside and outside of their polygons, and then the positional accuracy of each matched building pair is calculated as the distance between the centroids of corresponding building pairs in OSM and the reference data. If the matched pairs demonstrate a relationship of 1:1, the positional accuracy $\text{Accuracy}_{\text{posi}}$ is calculated as follows:

$$\text{Accuracy}_{\text{posi}} = \text{dis}(\text{CO}, \text{CF}) \quad (4)$$

where CO and CF refer to the centroids of the corresponding building footprints in OSM and the reference data, respectively. If the matched pairs demonstrate other relationships, such as 1: m , n :1 or n : m , then the centroids in formula (4) are replaced by the average centroids of the buildings as follows:

$$\text{CO} = \frac{1}{n} \sum_{i=0}^n \text{CO}_i \quad (5)$$

$$\text{CF} = \frac{1}{m} \sum_{j=0}^m \text{CF}_j \quad (6)$$

where CO_i and CF_j refer to the centroids of the building footprints in OSM and the reference data, respectively, and n and m describe the matched building-footprint pairs with a relationship of n : m .

3.3. Shape accuracy

This variable is a description of the similarity between a building footprint in OSM and that within the reference data set. The corresponding polygonal pairs in two data sets with relationships of 1:1 or n : m (including 1: m and n :1) are treated correspondingly when calculating the shape accuracy. The farthest point distance (FPD) is utilized as a shape signature to describe the shape, and a discrete Fourier transform is applied to calculate the shape accuracy (Xu *et al.* 2017) when the polygon pairs are characterized by a 1:1 relationship. A Fourier transformation describes the shape based on the fact

that any periodic signal of a finite extension can be expressed in terms of complex exponentials. Furthermore, a periodic signature function can be generated by a point moving around the boundary. The Fourier transformation of a boundary signature function generates a complete set of complex numbers known as the Fourier descriptors. In this way, the boundary can be expressed within the frequency domain, and the general shape is described by lower frequency information while smaller details are described by higher frequency information. Assuming that $f(u)$ is a shape signature, the discrete Fourier transform is as follows:

$$a_n = \frac{1}{N} \sum_{u=0}^{N-1} f(u) e^{-j2\pi nu/N} \quad n = 0, 1, \dots, N-1 \quad (7)$$

where $a_n (n = 0, 1, \dots, N-1)$ are the Fourier shape descriptors, denoted by FD_n . The shape signature $f(u)$ here is FPD, which can be calculated as follows:

$$f(u) = \sqrt{[x(u) - x_0]^2 + [y(u) - y_0]^2} + \sqrt{[x_{fp}(u) - x_0]^2 + [y_{fp}(u) - y_0]^2} \quad (8)$$

where $(x_{fp}(u), y_{fp}(u))$ is the farthest point from $(x(u), y(u))$ and (x_0, y_0) is the centroid of the shape. The shape accuracy is then calculated as follows:

$$\text{Accuracy}_{\text{shape}}(O, F) = 1 - \sqrt{\sum_{i=1}^M (f_i^o - f_i^f)^2} \quad (9)$$

where $\text{Accuracy}_{\text{shape}}$ is the shape accuracy, M is the index of the normalized Fourier descriptors, and f_i^o and f_i^f are the normalized Fourier descriptors for shapes O and F , respectively, which are the corresponding building footprints between that in the OSM and the reference data set.

If the matched pairs are related by $n:m$, a building may correspond to two or more touched (Figure 1(b)) or untouched (Figure 4(a)) footprints. Because the building footprints in OSM are digitized from Bing map imagery, separated buildings are combined to compare their shape similarity, and untouched footprints will be moved to meet with another one (Figure 4(b); small area polygons move towards larger area polygons) so that the dissolved buildings are treated as the corresponding pairs to calculate the shape accuracy. Throughout the majority of the map, whether the polygons belong to the same building in the real world cannot be determined without additional information because these polygons are physically separated on the map. This paper treats these separated polygons as one building only when they match with the same polygon in the other data set (i.e. Figure 3(a)).

3.4. Orientation consistency

This variable is utilized to assess the consistency of the orientation of a building, which is defined as the orientation of the minimum area-bounding rectangle (An 2011) corresponding to the building footprint in the map (Figure 5), and the length of the longer edge of the rectangle represents the size of the orientation vector. In the presented work, the relationship of the corresponding pair in the two data sets is first discovered. If

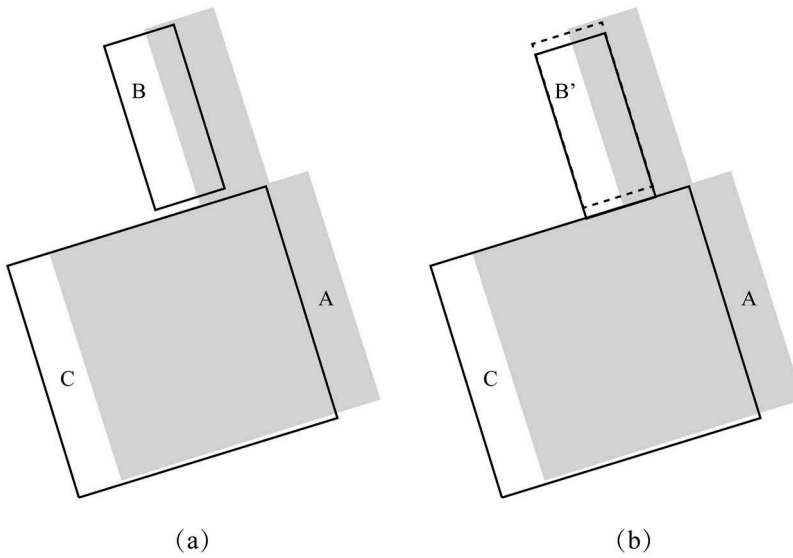


Figure 4. Untouched matched building pairs.

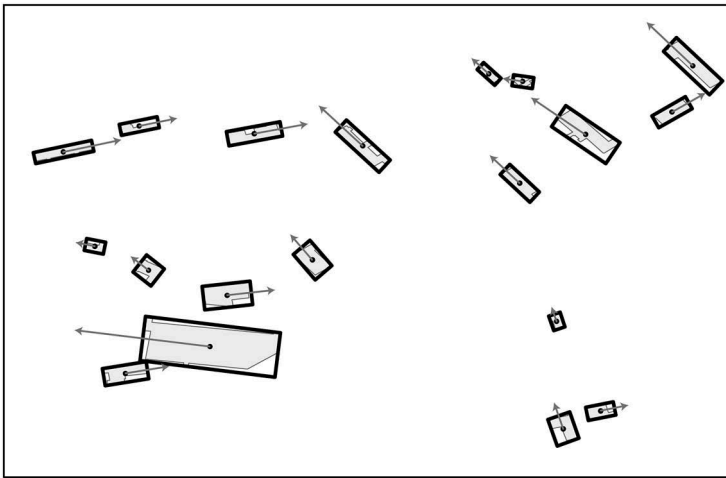


Figure 5. Direction of building footprints.

the pair of polygons demonstrates a 1:1 relationship, then the orientation consistency will be calculated one-by-one and calculated as follows:

$$\text{Consistency}_{\text{dire}} = 1 - \frac{|D_{\text{ref}} - D_{\text{OSM}}|}{180^\circ} \quad (10)$$

where D_{ref} and D_{OSM} refer to the orientation of a matched building footprint pair in the OSM and the reference data, respectively.

Otherwise, the orientation vector for the matched building-footprint pairs demonstrating an $n:m$ relationship (include $1:m$ and $n:1$) must to be calculated first, and the footprints belonging to the same building are then defined as a group. The orientation

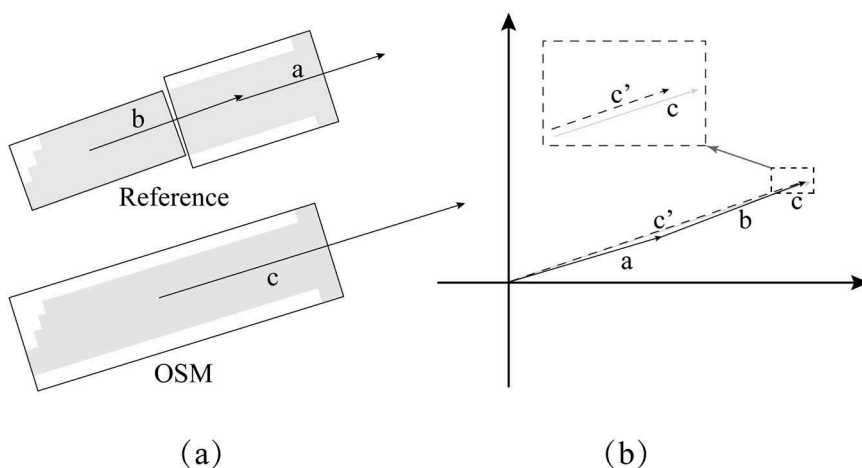


Figure 6. Vectors corresponding to the building direction.

vectors in the same group are added together to compare the group orientation with the orientation vector in the other data set (Figure 6). The orientation vectors in formula (10) are replaced by the sum of the grouped vectors as follows:

$$D_{OSM} = \sum_{i=0}^n D_i \quad (11)$$

$$D_{ref} = \sum_{j=0}^m D_j \quad (12)$$

where D_i and D_j refer to the orientation vectors of the building footprints in OSM and the reference data, respectively, and n and m describe the matched building-footprint pairs with an $n:m$ relationship.

The two vectors are added according to the triangle law: the vectors are first linked end-to-end, and then the summed vector is equal to the resultant vector that starts from the first vector's start point and ends with the second vector's end point (Figure 6 (b)). If there are more than two vectors in a group are needed to be added together, the sum of the first two is treated as a new vector, which is then added with the third; in this manner, all the vectors in the same group are added collectively as a final resultant vector.

3.5. Semantic accuracy

This variable defines what the object is represents an investigation into whether a building in the real world has been recorded within OSM. The semantic accuracy, in a broad sense, is equal to 100% (Fan *et al.* 2014). The semantic hierarchy and geometrical hierarchy have a closed relationship, and based on the OGC standard of CityGML building models (Gröger *et al.* 2008), the semantic accuracy is defined as follows: a building is recorded with correct semantics if it demonstrates a 1:1 relationship with the

reference data, and if one building corresponding to m buildings in the reference (1: m) shows that it is of a higher-level semantic hierarchy to the corresponding buildings. A 1:0 relationship represents a building in OSM that has no matched pairs in reality (e.g. semantically incorrect), whereas a 0:1 relationship represents an opposite case. A building in OSM is a part of a building in the reference data (n :1); in other words, the building is recorded at a low level on the semantic hierarchy. Matched pairs with an n : m relationship also express an incorrect record with respect to semantic accuracy. Thus, as previously proposed (Fan *et al.* 2014), the semantic accuracy is calculated as follows:

$$\text{Accuracy}_{\text{semantic}} = \frac{N_1 + N_2}{N} \quad (13)$$

where N_1 and N_2 refer to the number of buildings in OSM that match the reference data set in relationships of 1:1 and 1:0 and N stands for the number of building footprints in the OSM. OSM data are frequently updated; thus, based on a visual inspection of Bing map and Google map imagery, the OSM buildings that match those in the reference data set in a relationship of 1:0 are added during calculating the semantic accuracy in formula (13).

4. Deep autoencoder network quality assessment

The quality assessment of OSM data for building footprints includes the proposed defined measures of data quality, establishing the metrics for measuring the measures and finally communicating the data quality (Fisher *et al.* 2006). The method of quality assessment by the deep autoencoder network involves three steps. First, data from OSM and the reference data set are divided into cells and matching corresponding building footprints. Second, the aforementioned assessment measures are compared. Third, the deep autoencoder network is trained using the metric measures until the reconstruction errors are minimized and stable. The quality of OSM is assessed by the difference between the input and output of the deep network.

4.1. Deep autoencoder network

The autoencoder network proposed by Hinton and Salakhutdinov (2006) is considered as a new model of a neural network, and it contains many layers to encode and decode the input data. The deep autoencoder network has a symmetrical structure that is trained to minimize the differences between the input and output. The output data are estimates of the input data, and the difference between the two is defined as the reconstruction error, which is utilized to detect anomalies (Fiore *et al.* 2013, Sun *et al.* 2014). A larger error corresponds to more anomalous input data, and the reconstruction error is calculated as follows:

$$E = \sqrt{\sum_{i=1}^n (O_i - I_i)^2} \quad (14)$$

where E represents the reconstruction error, O represents the output data, I represents the input data and n depicts the size of the output layer.

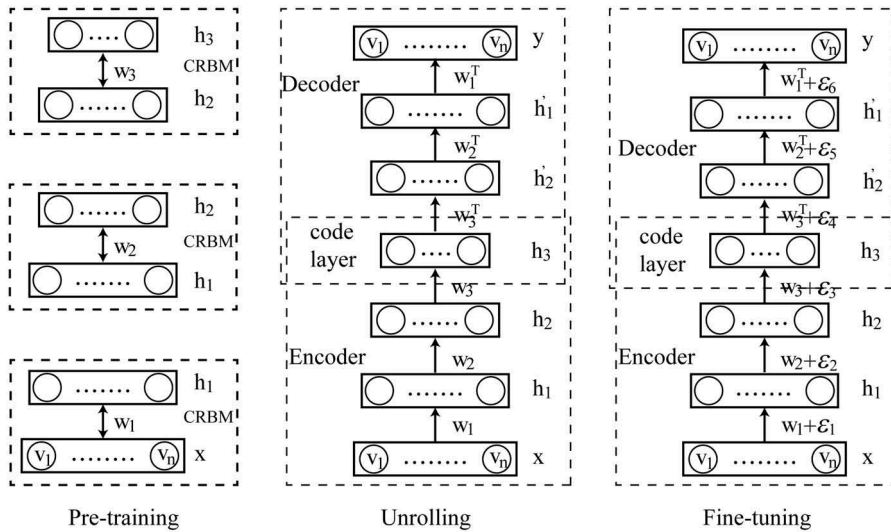


Figure 7. Unrolling of the deep autoencoder network.

The deep autoencoder network is trained to minimize the reconstruction error through three steps. First, in the pre-training step, CRBMs are trained one by one. Then, in the unrolling step, all of the trained CRBMs are stacked to form a deep autoencoder network. Finally, a back-propagation algorithm is introduced to adjust all the parameters in the network. This step is known as fine-tuning (Figure 7).

During the training, the activation function of an autoencoder network is defined as follows:

$$\varphi(x_j) = \theta_L + (\theta_H - \theta_L) \frac{1}{1 + \exp(-a_j x_j)} \quad (15)$$

where $\varphi(x)$ is a sigmoid function whose asymptotes are the constants θ_L and θ_H . The parameter a_j controls the slope of the sigmoid function and ensures a smooth transition from the deterministic state without noise to the binary-stochastic state while a_j increases (Chen and Murray 2003).

The CRBM is trained iteratively using the MCD rule in this paper, and back-propagation is introduced to tune the parameters in the network by evaluating the error in every unit and calculating the contribution of the unit to the reconstruction errors. The parameters are tuned by minimizing the reconstruction errors.

4.2. Quality assessment using the deep autoencoder network

We can obtain a multitude of cells by partitioning the OSM and reference data set. A cell in OSM corresponds to only one cell in the reference data set, and they contain the same ID. The cells in both data sets with the same ID are treated as a data sample to calculate the measures for building footprint quality assessment. Consequently, the calculated results of measures for a pair of cells are utilized as a data sample for the deep autoencoder network.

After pre-training and fine-tuning, once several parameters for the network and the CRBMs have been established and the deep autoencoder network has been stable, then the data sample can be reconstructed. Every set of measures for a pair of cells as input data corresponds to a reconstruction result obtained through the deep autoencoder network, and the errors between the input data and the reconstructed results represent the OSM data quality performance. Each input of the cell relates to an error, and a larger error indicates a more anomalous cell performance. An anomalous performance may be positive or negative. If the input corresponds to a larger error and exceeds the mean value, then we regard the cell as demonstrating a better quality. Otherwise, the quality of the cell is relatively poor.

5. Experiments and discussion

Both official and OSM data for Toronto, Canada, were utilized to test the proposed method, and the official data set was treated as the reference data in the experiment. The reference data, which was collected in 2014, were prepared by the City Planning and the Municipal Property Assessment Corporation to a building accuracy of ± 1 m. In addition, the OSM data set was collected in 2015 and exhibits an average offset of 4.13 m, which corresponds to the precision of Bing map imagery is estimated at 3–4 m (Fan *et al.* 2014). Toronto is located in the southernmost extent of Canada. It is the largest city in the country and has more than 400,300 building footprints in the reference map. The OSM and the reference map show identical boundaries for the city of Toronto. However, the number of building footprints in OSM is only approximately 40,000 (Figure 8). In this section, we discuss how to train a deep autoencoder network and set the requisite parameters. In addition, different data division methods were compared according to the quality assessment results.

5.1. Data preprocessing and calculation of assessment measures

In practice, adjacent buildings (e.g. buildings that share common boundaries) may be described as single objects in a detailed expression map. However, if the map is an approximation, some adjacent buildings may be expressed as a synthesized object. Consequently, the occurrence of adjacent buildings is obvious from the difference in the number of footprints in the OSM and reference map, which are matched not only with a 1:1 relation but also with a $m:1$ relationship. Figure 9 illustrates the proportion of the relationship types between the OSM and reference data set.

Before calculating the measures, a cellular grid is initiated over Toronto on both the OSM and reference map. Different sizes and styles of cells may lead to different results. Using a geometrical method, the map can be divided into squares or hexagons. Hence, in this study, we divided the map into square cells with side lengths of 200 m, 100 m and into hexagonal cells of length 150 m as the method mentioned in Section 2.2. The results obtained using the different cell styles were compared at the end of the experiment.

The cells in the OSM and reference maps that were split by the same templated grid were labelled with the same IDs, which was then utilized to compute the assessment measures. The OSM and reference data sets were divided and matched according to the

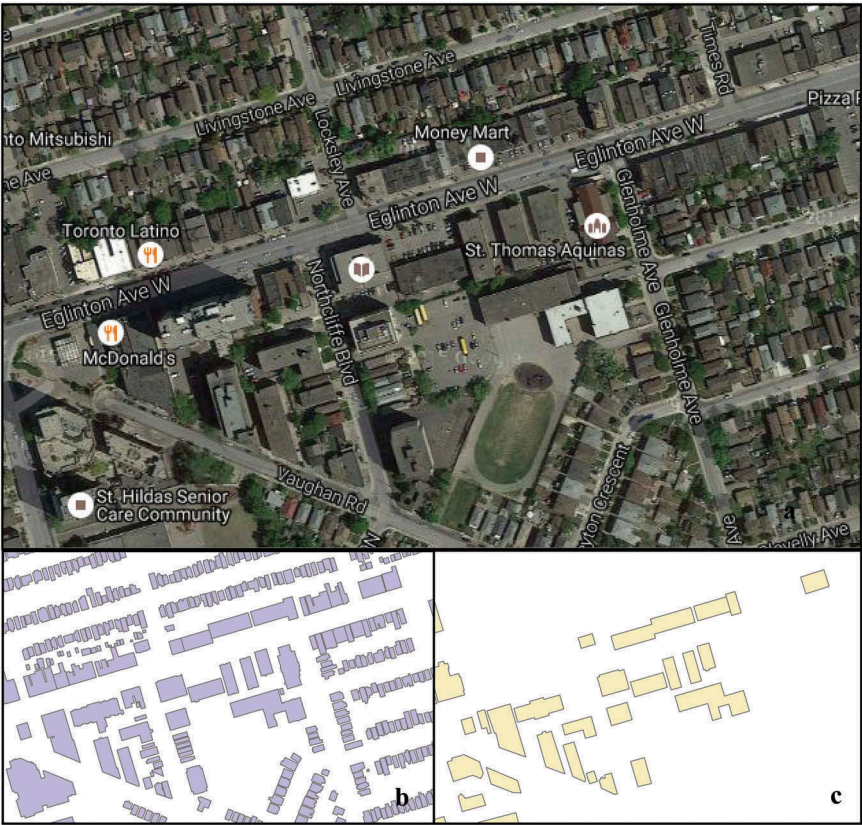


Figure 8. Small region of Toronto in Google Earth (a), the reference data set (b) and the OSM data set (c).

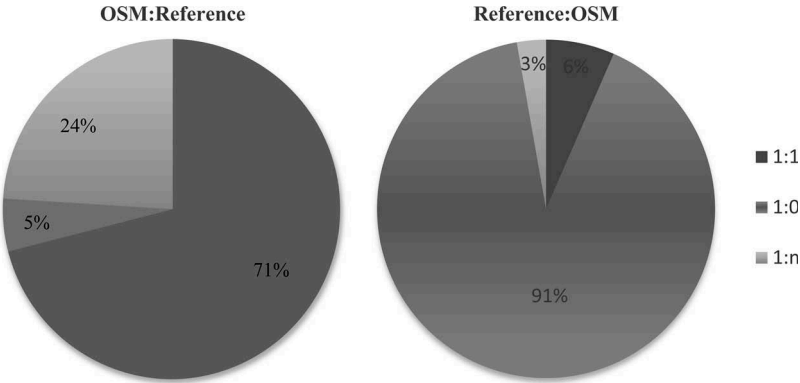


Figure 9. Proportion of relationships between the OSM and reference data set.

method presented in [Section 2](#). Assessment indices were calculated within each cell, and there are 22,049, 8,634 and 10,412 cells covering the study area when the map is divided using the 200 m and 100 m square cells and the 150 m hexagonal cells, respectively.

Subsequently, the proposed measures of data completeness, positional accuracy, shape accuracy, orientation consistency and semantic accuracy were calculated cell-by-cell employing the methods described in [Section 3](#). During the calculation of the positional accuracy, the accuracy was normalized to [0, 1]. In addition, the shape accuracy describes the mean similarity of corresponding building footprints within each cell from different data sets; thus, the similarity between every matched pairs must be computed first.

In this manner, every pair of cells was evaluated via the five elements described in [Section 3](#), and the pair is treated as a sample point, where the shape accuracy is denoted along with the average values of the individual footprints within the cell, and positional accuracy and orientation consistency the same as shape accuracy. The number of samples equals to the number of cells, and every sample point contains five-dimensional data. For example, there are 22,049 samples when the map is divided by the 100 m $\times$ 100 m cells. In the experiment, those samples were used as the input data for the deep autoencoder network. After coding and decoding, the quality of the building footprints was assessed comprehensively.

5.2. Autoencoder network architecture selection and parameters setting

The structure is vital for any reliable network because it includes the depth of the autoencoder network as well as the number of nodes of each layer and other attributes. At the same time, there are many parameters such as the weights between two layers, biases in every layer and learning rates for the network. Proper initialization of these arguments in pre-training and fine-tuning is important to assure network performance.

At a given samples number, the autoencoder network may perform poorly when increasing hidden layers. In such a situation, a neuron is usually added so that the mathematical formulation of the model will have additional parameters, and it provides more degrees of freedom to fit the model, which may lead to network overfitting (Larochelle *et al.* 2009). Subject to these factors, three CRBMs are stacked in the autoencoder network in this study ([Figure 7](#)).

After the number of hidden layers within the deep neural network was set, the appropriate size of each layer shall be determined. If the hidden layer has fewer units (e.g. a smaller layer size), the model will have fewer resources with which to learn the training sample population features (Chen 2015). However, adding more hidden units means using more parameters that will more likely cause overfitting (Hinton 2012). In this paper, we compared the results from three different general cases wherein the hidden layer size either increases (doubles), decreases (halves) or remains unchanged while the number of layers increases ([Figure 10](#)). The deep network performs the best when the size of its hidden layers decreases as the number of its layers increases. Based on the above discussion and the experimental analyses ([Figure 10](#)), the number of the layer size was set to 70. Because the deep autoencoder network shows a symmetrical structure, a 40-20-10-20-40 autoencoder network is established experimentally.

Before training the network, some parameters must be initialized, such as the weights, biases and the momentum that is used to increase the learning speed. Small values from a zero-mean Gaussian with a standard deviation of 0.01 were selected as

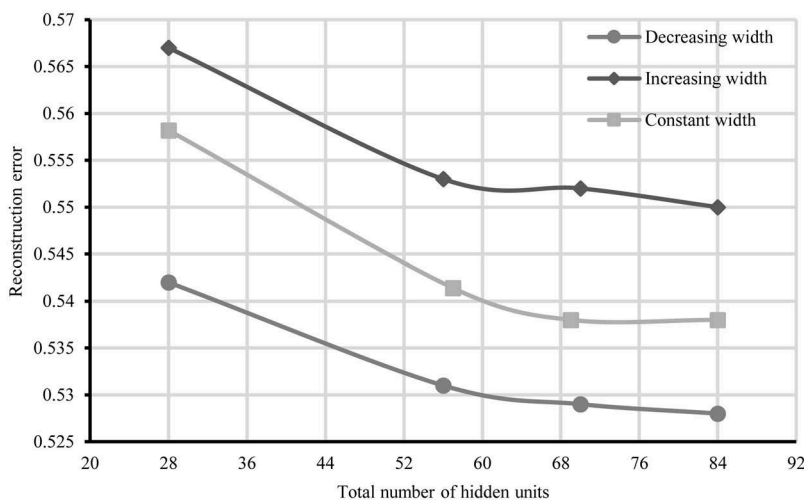


Figure 10. Reconstruction error changes of three-layer deep networks for three types of structures (increase, decrease and constant).

weights, and hidden biases were set to 0 (Hinton 2012). The parameter utilized to control the slope of the sigmoid function in formula (15) is set as $a_j = 0.1$. In addition, at the start of the learning, random initial parameter values may lead to a large gradient; therefore, setting a low momentum is preferable. However, once the reconstruction errors have settled down to gentle progress, the momentum should be increased (Hinton 2012). As recommended by Hinton (2012), the momentum was initialized with a value of 0.5, and it increased to 0.9 when the reconstruction error settled down.

Setting learning rate during pre-training and fine-tuning plays an important role when training the model because the rate controls the size of the increment while the weights and bias are being updated. An oversized learning rate may lead to dramatic changes in the reconstruction error and considerable increases in the weight values. In addition, the reconstruction errors usually fall while reducing the learning rate (Hinton 2012). However, if the learning rate keeps increasing when it is already sufficiently large, then the trained model may become overfitting (Valentine and Trampert 2012). Based on our analysis (Figure 11(a)), the learning rate during pre-training for CRBM was set to 0.15. We set an appropriate learning rate as 5 in fine-tuning in the same way (Figure 11(b)). Small differences in the weight cost are unlikely to cause big differences in performance (Hinton 2012). As recommended by Hinton (2012), weight cost was set to 0.0001 during training the CRBM, which is derivative of the penalty term by the learning rate to optimize the weights changing.

To ensure that the threshold T_0 in formulas (1) and (2) is appropriate, this paper randomly selected a number of corresponding buildings from OSM and the reference data set, respectively, and analysed how the completeness of the matched buildings was affected by the threshold. An improper threshold may lead to a mismatch of a building in one data set with its corresponding neighbour or the inability to locate a match in the other data set. After the sensitivity analysis (Figure 12), the threshold was set to 30% in

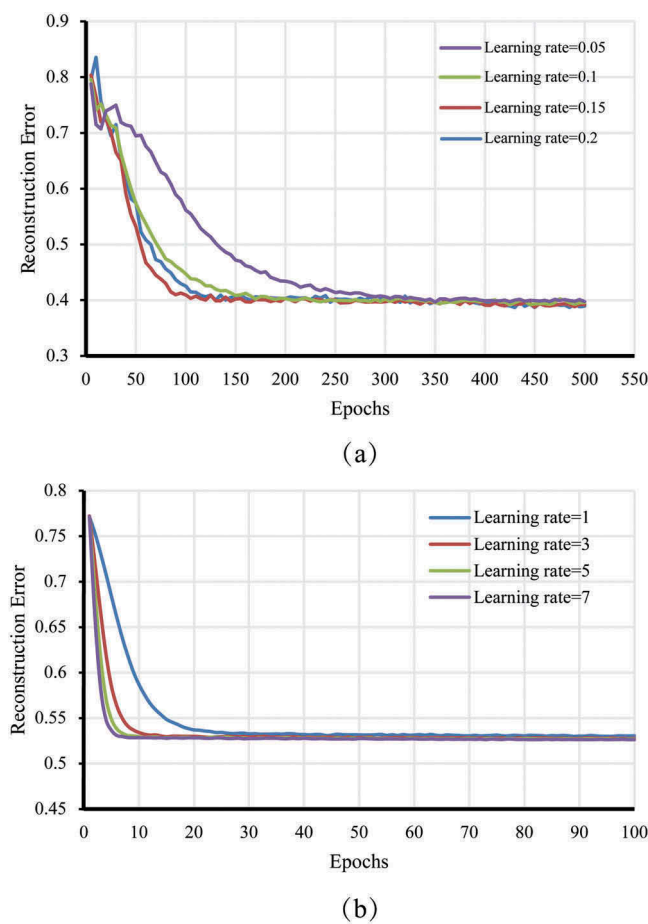


Figure 11. Learning rate for the first CRBM during (a) pre-training and (b) fine-tuning.

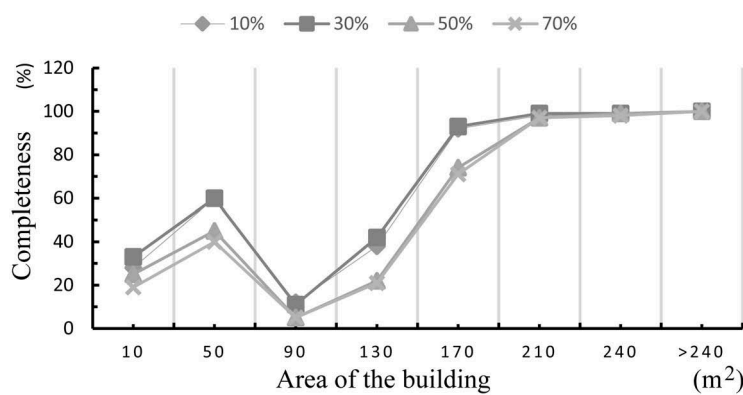


Figure 12. Completeness variations affected by the threshold value.

the experiment. Figure 12 shows that the completeness is always lower when the building is smaller because some smaller buildings in denser districts are hardly digitized and usually don't appear in OSM as shown in Figure 8.

5.3. Quality assessment and analysis

The composite index for all the proposed measures ranges $[0, 1]$; therefore, θ_L and θ_H were set as 0 and 1 in formula (15), respectively. After the initiation of the architecture and parameters of the deep autoencoder network, the network was trained through 500 iterations. Subsequently, we encoded the samples with the trained network, and the normalized reconstruction error for every sample was used to assess the quality of OSM building footprints (Figure 13), wherein every cell was treated as a sample data point.

The method discussed in this paper is based on a cellular with a scale and type that may lead to different results: with small cell division providing more detailed assessment results than larger division and hexagonal cells producing relatively smoother results than square cells. For verification, we compared the cells divided using different squares and hexagons and discovered consistent quality distributions in all cases (Figure 13). However, the $100\text{ m} \times 100\text{ m}$ square cell (Figure 13(b)) resulted in a finer quality distribution compared with the output based on $200\text{ m} \times 200\text{ m}$ square cell (Figure 13(a)) because in the former an area of $10,000\text{ m}^2$ was treated as a sample when assessing the quality, whereas in the latter an area of $40,000\text{ m}^2$ was treated as a sample. Hexagons is widely applied to tessellate maps because they can be assembled into an earth-like sphere and have larger area than a square with the same side lengths,

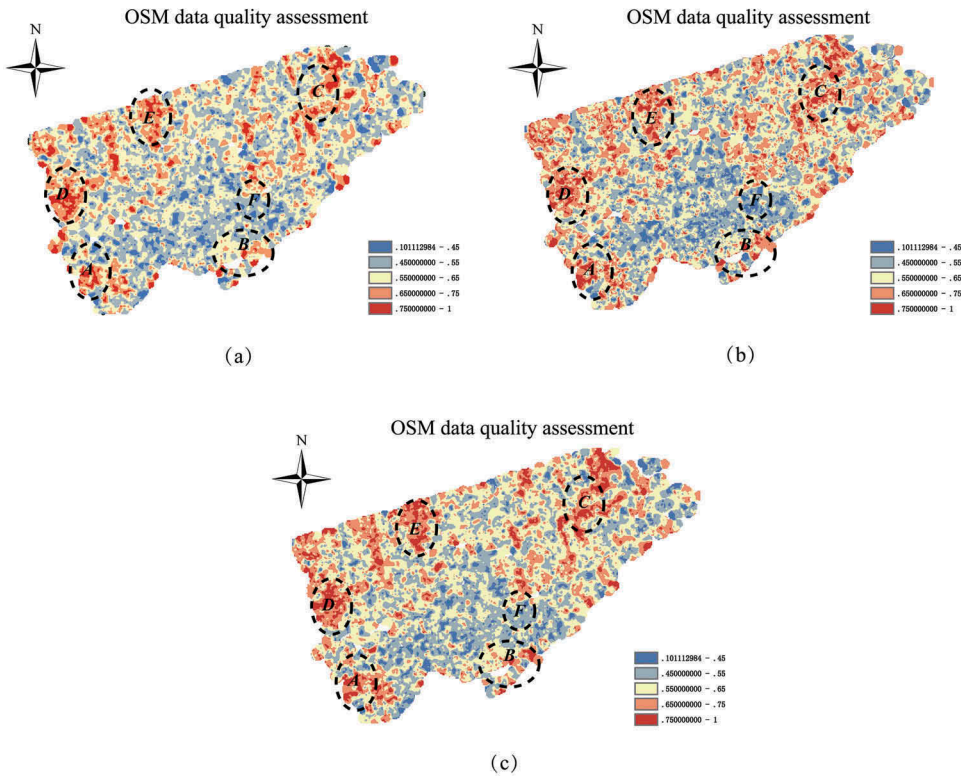


Figure 13. OSM building footprint quality assessed in different grids: (a) $200\text{ m} \times 200\text{ m}$ square, (b) $100\text{ m} \times 100\text{ m}$ square and (c) hexagon.

which collectively benefit the continuity of the results (Figure 13(c)). Accordingly, hexagons have more neighbours (edges); therefore, this shape will produce a smoother assessment result than square cells.

OSM data quality is closely related to a series of factors, such as the environmental impact, population density, data collected devices and other factors (Zheng and Zheng 2014). The OSM data quality distribution is a consequence of many contributors providing data for a developed district than for a developing region (Dorn *et al.* 2015). This postulation is supported by an examination of the evolution of Toronto as represented by all of the cell configuration cases (Figure 13). For instance, the quality of the city halls in Figure 13, such as these in Etobicoke (A), Toronto (B) and Scarborough (C), as well as the district near the Toronto Congress Centre (D) is relatively high, whereas the quality for other suburbs is poor. Moreover, the OSM building footprint quality map (Figure 13) corresponds with the population distribution. Expert or trained contributors of VGI will generally provide high-quality volunteered data (Neis and Zielstra 2014, Hung *et al.* 2016). The district near York University (E) clearly demonstrates an excellent performance on the map. The geographical environment is also closely related to OSM data quality, and the results assessed by the deep autoencoder network shows a poor-quality region near the Lower Don Parklands (F) because of the mountains and plants.

To validate the training outcome and the generalization of the training model, we separated a data set (200 m × 200 m 8634 samples) into a training data set (6907 samples) and a test data set (1727 samples). The test results (Figure 14) indicated that the training outcome is consistent with the assessment results in Figure (13(a), A and B), although certain parts (e.g. C) demonstrate slight differences that are caused by the number of trained samples. Hence, the model can be generalizable to assess data quality.

To explain the assessment results in terms of the different quality assessment measures, this paper assessed the data quality by the measures alone, and using the deep autoencoder network absent a selected single measure (Figure 15). Our findings indicate that completeness provides the most significant contribution to the results because the completeness of VGI data has regional characteristics of the quality distribution. However, shape similarity contributes the least because most of the buildings are simple polygons and the differences between them are minor. Most of the buildings have decent orientation consistency, which helps improve the results in certain areas. The positional accuracy and semantic accuracy have a certain randomness, which helps to adjust the assessment results in detail.

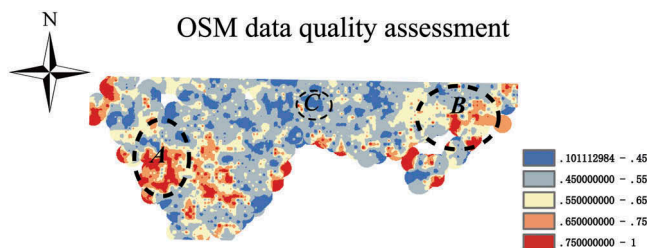


Figure 14. Test data for validating the model.

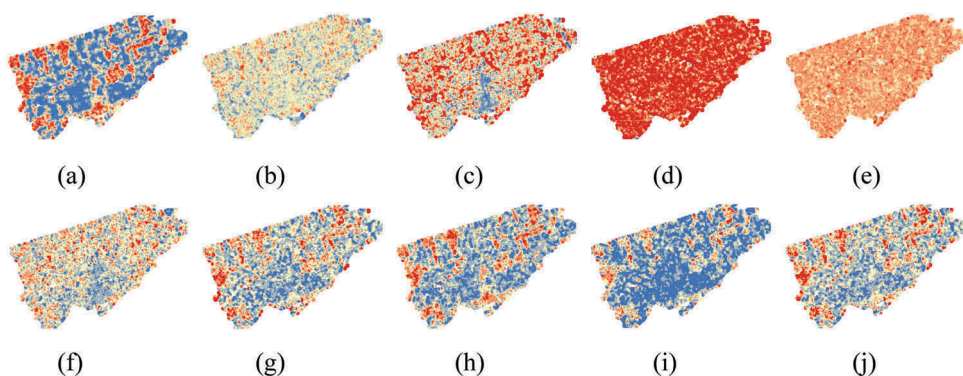


Figure 15. Effects of the assessment elements. Parts (a)–(e) demonstrate completeness, positional accuracy, semantic accuracy, directional consistency and shape similarity values; and parts (f)–(j) correspond to the results assessed without the respective elements.

The utility of the different methods can be comprehensively and objectively assessed through an analysis of the experimental results, such as for two cells in region (A) in [Figure 13\(a\)](#), in which five assessment metrics presented in [Section 3](#) are (1, 0.5048, 0.6726, 0.952, 1) and (1, 0.9204, 0.6011, 0.8494, 1) for the respective cells. Compared with other methods, this work can assess the quality of the first cell is 0.7339, whereas that of the second is 0.8297. Moreover, the method employed in this study introduces orientation consistency to assess data quality because it is useful for analysing circumstances in which one building footprint faces north (0°) in the reference while its corresponding footprint faces northeast (5°) in OSM. In such circumstances, other available methods may have limited effectiveness. During experimental matching, certain cases may be similar to [Figure 1](#), wherein only an overlap will lead to a mismatch, and thus, corresponding buildings are matched by filtering and refinement in this work.

6. Conclusions

Quality assessments are very important for user-generated spatial data if these data are used to meet specific needs or fulfil certain requirements. This paper employs a deep autoencoder network to analyse numerous accuracy measures, including data completeness, positional accuracy, shape accuracy, orientation consistency and semantic accuracy, and to obtain a more comprehensive and objective quality assessment result than that can be obtained using other methods. For calculating the shape accuracy, positional accuracy and orientation consistency, this work considers matched pairs with the relationships of not only 1:1 but also 1: m , n :1 and n : m , unlike other OSM quality assessment methods. By calculating the shape accuracy using Fourier descriptors, the method does not need to consider start or end point. To consider the rotation of the building footprints, this method introduces the orientation consistency to properly assess the data quality. Moreover, the method for matching building footprints has been improved, which tunes the corresponding relationships.

We utilized the proposed assessment model to evaluate OSM data for Toronto. The experiments analysed the assessment results in terms of the five quality assessment

measures (Figure 15) as well as the manner in which the spatial configuration network affects the results (Figure 13). The experimental results were obtained by analysing the deep level data law instead of by summing the measures with weights provided by experts or experienced researchers, thereby providing a comprehensive and objective assessment result. OSM is a popular crowdsourced data repository that depends on the status of development of the region, the density of the populace, etc. These factors are closely related to the OSM data quality and clearly correspond to the assessment results (Figure 13), as illustrated by the quality distribution map.

An optimal network is determined by many factors, such as the operational parameters and network architecture. In this paper, these parameters, such as the weights and biases, are initialized using a zero-mean Gaussian function with a standard deviation, as recommended by previous studies. The learning rate, network architecture and the number of iterations were designed by observing the reconstruction errors. We believe that the learning rate, network architecture and iteration number are optimal when the reconstruction errors are minimized and stable. Moreover, the autoencoder used in this paper learns the data features based on the probability, and every measure of a data point selected as the input of the network for assessing the data quality should be independent. Therefore, we have introduced five assessment measures, which are described in Section 3.

Acknowledgement

The authors thank Prof. May Yuan, Bo Huang and two anonymous reviewers for their critical reviews and constructive comments that improved the manuscript. This work was supported by the National Natural Science Foundation of China (No: 41,671,400, No: 41,401,443), Hubei Natural Science Foundation of China (No: 2015CFA012), Research Funds for the Central Universities basic special projects (No. CUG160226).

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by the National Natural Science Foundation of China [(No. 41401443, No. 41671400); Hubei Natural Science Foundation of China [No: 2015CFA012]; Fundamental Research Funds for the Central Universities, China University of Geosciences (Wuhan) [No. CUG160226].

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